Master's Thesis Progress Report

Friday, July 10th

Focus:

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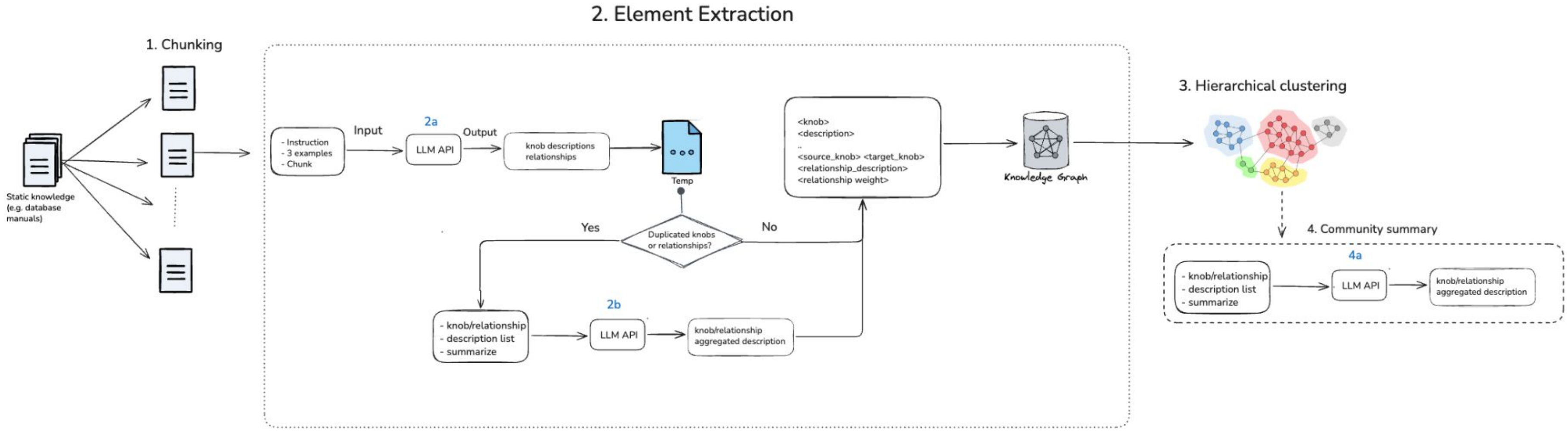
Total: 11 Slides

Component 1: Knowledge Base construction



Input data preparation

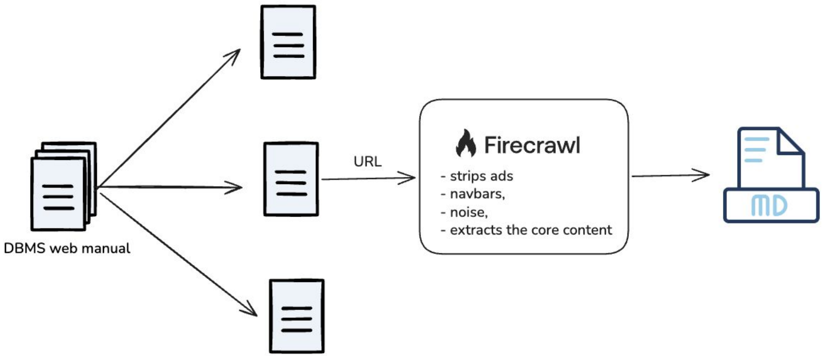
DBMS manual:

1) PDF
    - RABBIT does not take input as PDF files
    - PDF to text conversion is challenging
2) Web
    - HTML tags noise
    - Utilize existing tools to extract core text

Select 3 pages:

- InnoDB parameters
- Internal Temporary Table
- What the MySQL Upgrade Process Upgrades

Total: 184 knobs

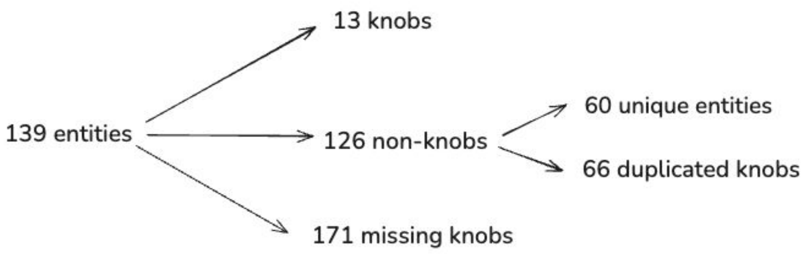


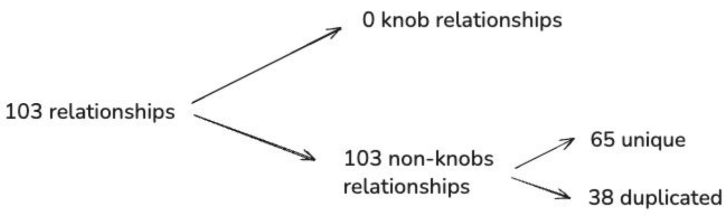
2. Element extraction - LLM 2a

Goal: Extract from chunks

- All knobs + descriptions
- All knob relationships + description

Hallucination type: Entity and relationship error





Precision and Recall

	Knob extraction	Relationship extraction
Precision	0.17	0
Recall	0.07	0
F1 Score	0.10	0

Solution

Focus on LLM-2a component where most of the hallucination happens

Cause of hallucination:

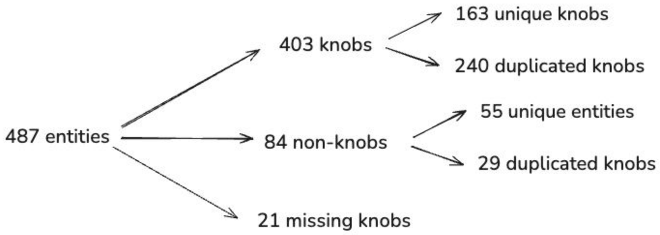
- Default GraphRAG prompt
- Default entity types

Prompt Template

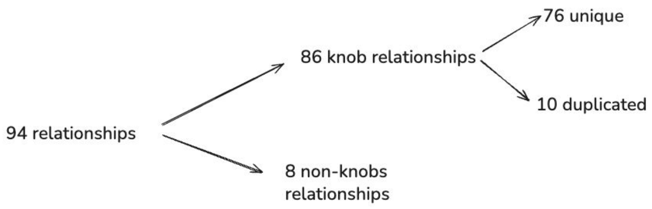
```
-Goal-
Given a text document that is potentially relevant to this activity, identify the knobs and the relatinoships among the
identified knobs.

Database: {dbms_name} {dbms_version}

Knob definition: {knob_definition}

DBMS-specific terminology:
In {dbms_name}, knobs may be referred to as: {knob_terms}.

-Steps-
1. Identify all the knobs. For each knob, extract the following information:
- knob_name: Name of the knob, capitalized
- knob_description: Description of the knob in 1-3 sentences. Include any default values or ranges if mentioned in the text.
If description is not included, keep it empty.
Format each knob as ("knob"{tuple_delimiter}<knob_name>{tuple_delimiter}<knob_description>)

2. From the knobs identified in step 1, identify all pairs of (source_knob, target_knob) that are *clearly related* to each
other.
For each pair of related knobs, extract the following information:
- source_knob: name of the source knob, as identified in step 1
- target_knob: name of the target knob, as identified in step 1
- relationship_description: explanation as to why you think the source knob and the target knob are related to each other
 Format each relationship as ("relationship"{tuple_delimiter}<source_knob>{tuple_delimiter}<target_knob>{tuple_delimiter}
 <relationship_description>)

3. Return output in English as a single list of all the knobs and relationships identified in steps 1 and 2. Use **
{record_delimiter}** as the list delimiter.

4. When finished, output {completion_delimiter}

######################
-Examples-
######################
{examples}

######################
-Real Data-
######################
Text: {input_text}
######################
Output:
```

2. My solution: Element extraction - LLM 2a

Knobs



Relationships



My solution: Element extraction - LLM 2a

Knobs

- Precision: **0.74** (0.17)
- Recall: **0.88** (0.07)
- F1-score: **0.80** (0.10)

Relationships

- Precision: **0.9** **(0.75)**
- Recall: **NA** **(0.00)**
- F1-score: **NA** **(0.00)**

Next steps

01 Count the number of relationships in the input file

02 Calculate Recall and F1 score for my solution

02 Come up with a solution to filter the input data

Appendix

68 unique knobs in input

78 knobs

11 duplicated knobs

11 Correct knob relationships

11 relationships

22 missing relationships

True Positive (TP)	68	Extracted entity is one of the 68 input knobs
False Positive (FP)	0	Extracted entity is not one of the 68 input knobs
False Negative (FN)	0	Input knob was missed by RABBIT

True Positive (TP)	11	Extracted relationship is a correct knob dependency
False Positive (FP)	0	Extracted relationship is not a correct knob dependency
False Negative (FN)	22	A real dependency exists in the input, but RABBIT missed it